AWS VPC PEERING PROJECT REPORT

EWB Courses – AWS Networking Lab

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Project Title: Secure Private Communication Between Two Amazon VPCs Using VPC Peering
AWS Region: Asia Pacific (Sydney) – ap-southeast-2
Completion Date: 27 October 2026

1. Introduction

Amazon Virtual Private Cloud (VPC) allows users to create isolated virtual networks in AWS. VPC Peering enables communication between two VPCs using private IP addresses. The objective of this project is to establish private communication between two VPCs and verify connectivity using SSH and ping commands.

2. Project Architecture

This project contains two VPCs: VPC A and VPC B. VPC A contains a public subnet with a bastion host and a private subnet with App-A. VPC B contains a private subnet with App-B. The VPCs are connected through an active VPC peering connection.

2.1 Resource Inventory

Resource	Resource ID / Name	CIDR or Private IP
VPC A	ewb-vpc-a	10.10.0.0/16
VPC A public subnet	ewb-subnet-a-public	10.10.1.0/24
VPC A private subnet	ewb-subnet-a-private	10.10.2.0/24
VPC B	ewb-vpc-b	10.20.0.0/16
VPC B private subnet	ewb-subnet-b-private	10.20.1.0/24
Bastion host	ewb-bastion	10.10.1.192
App-A	ewb-app-a	10.10.2.10
App-B	ewb-app-b	10.20.1.10
Peering connection	ewb-a-b-peer	pcx-0a97b3dbb6a1ae556

3. Route Table Configuration

The VPC A public route table contains a local route for 10.10.0.0/16, a default route to the Internet Gateway, and a route to 10.20.0.0/16 through the peering connection.

The VPC A private route table contains a local route for 10.10.0.0/16 and a route for 10.20.0.0/16 through the peering connection.

The VPC B private route table contains a local route for 10.20.0.0/16 and a route for 10.10.0.0/16 through the peering connection.

These routes allow traffic to travel between both VPCs through private IP addresses.

4. Security Configuration

The bastion host is used as a controlled entry point to the private instances. App-A and App-B do not have public IPv4 addresses. Security groups control SSH access and ICMP traffic for ping testing.

SSH access is permitted from the bastion to the private instances. The required ICMP rules allow the two application instances to test connectivity. Security group rules should allow only the necessary sources and protocols.

5. Testing and Results

5.1 SSH Connection

SSH access from the local computer to the bastion host was tested. From the bastion host, access to App-A was successful. SSH access from App-A to App-B was also successful after configuring the required security group rule.

5.2 App-A to App-B Ping Test

The command ping -c 4 10.20.1.10 was executed from App-A. Four packets were received successfully, with 0% packet loss.

5.3 App-B to App-A Ping Test

The command ping -c 4 10.10.2.10 was executed from App-B. Four packets were received successfully, with 0% packet loss.

These results demonstrate successful bidirectional private communication between the application instances.

6. Network Traffic Flow

When App-A sends a packet to App-B, the route table in VPC A forwards the packet for the 10.20.0.0/16 network through the active peering connection. VPC B then delivers the packet to App-B.

The reply packet follows the reverse route. VPC B forwards traffic destined for 10.10.0.0/16 through the peering connection, and VPC A delivers it to App-A. Both route tables and the relevant security group rules are required for successful communication.

7. Limitations and Additional Evidence

The supplied screenshots demonstrate an active peering connection, route table entries, and successful ping tests. Baseline failure before peering, controlled ICMP blocking and recovery, and cleanup evidence are not demonstrated in the supplied screenshots. These tests should be performed and documented if required by the course workbook.

8. Conclusion

The project successfully demonstrated private communication between two Amazon VPCs using VPC Peering. The successful ping tests in both directions confirmed that the required routes and ICMP permissions were configured correctly during testing.

The project helped demonstrate the importance of route tables, security groups, private IP addressing, SSH access, and testing both forward and return traffic paths.

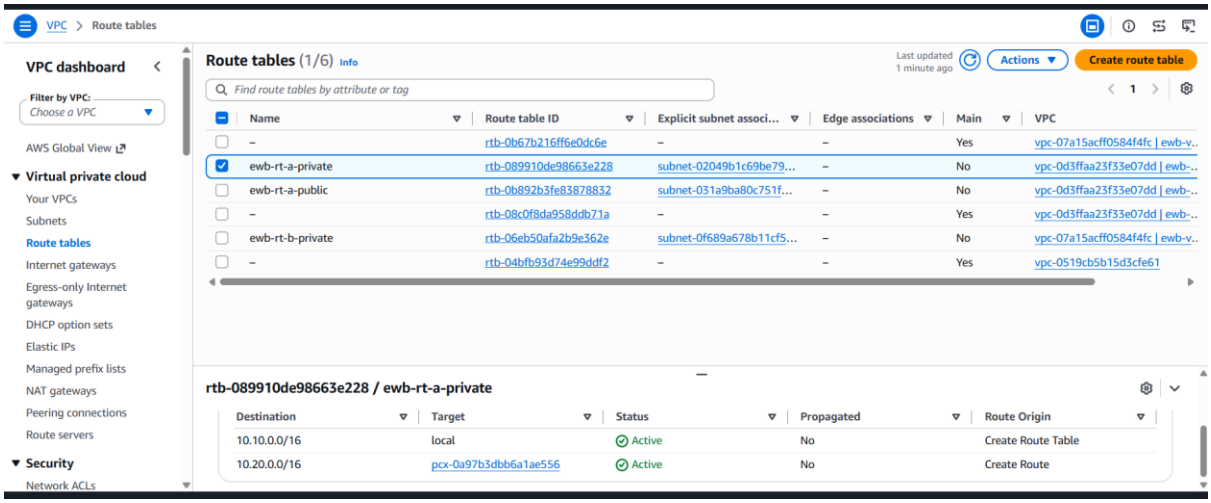
9. Security and Cleanup

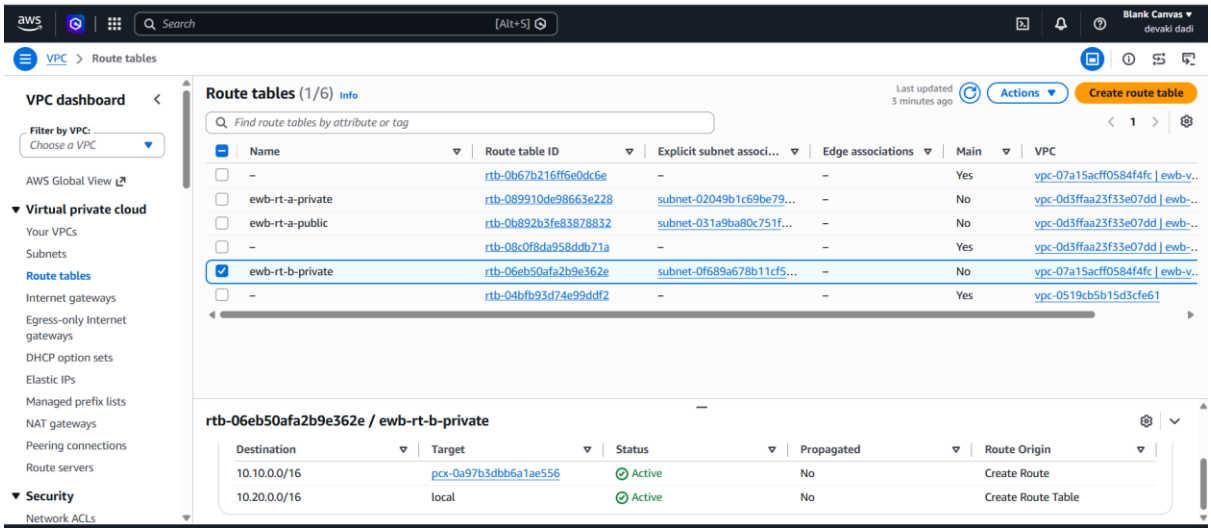
Private keys, passwords, AWS access keys, and other credentials must not be included in the report or uploaded to GitHub. After collecting the required evidence and completing submission, lab resources should be cleaned up according to the course instructions to avoid unnecessary AWS charges.

Appendix: Screenshot Evidence

Insert the screenshots showing:

1. Active VPC peering connection.
2. VPC A private route table.
3. VPC B private route table.
4. App-A to App-B successful ping.
5. App-B to App-A successful ping.
6. SSH connection to App-B.
7. Any additional required security group and cleanup evidence.






```
[ec2-user@ip-10-10-2-10 ~]$ ssh -A ec2 -user@10.10.2.10
unknown option -- u
usage: ssh [-46AaCfGgKkMNnqsTtVvXxYy] [-B bind_interface] [-b bind_address]
           [-c cipher_spec] [-D [bind_address:]port] [-E log_file]
           [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
           [-J destination] [-L address] [-l login_name] [-m mac_spec]
           [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
           [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
           destination [command [argument ...]]
       ssh [-Q query_option]
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.166 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.167 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.191 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.189 ms

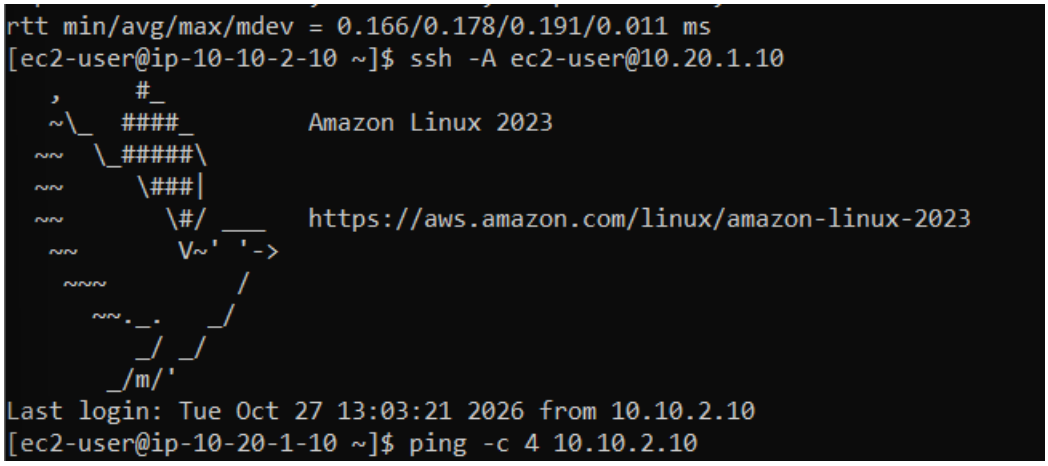
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3088ms
rtt min/avg/max/mdev = 0.166/0.178/0.191/0.011 ms
[ec2-user@ip-10-10-2-10 ~]$ _
```

```
tt min/avg/max/mdev = 0.166/0.178/0.191/0.011 ms
ec2-user@ip-10-10-2-10 ~]$ ssh -A ec2-user@10.20.1.10
  ,     #_
  ~\_  ####_        Amazon Linux 2023
 ~~  \_#####\
 ~~     \###|
 ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
  ~~       V~' '->
   ~~~         /
     ~~._.   _/
        _/ _/
      _/m/'
ast login: Tue Oct 27 13:03:21 2026 from 10.10.2.10
ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
ING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
4 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.137 ms
4 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.197 ms
4 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.177 ms
4 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.176 ms

-- 10.10.2.10 ping statistics ---
 packets transmitted, 4 received, 0% packet loss, time 3145ms
tt min/avg/max/mdev = 0.137/0.171/0.197/0.021 ms
ec2-user@ip-10-20-1-10 ~]$
```

```
rtt min/avg/max/mdev = 0.166/0.178/0.191/0.011 ms
[ec2-user@ip-10-10-2-10 ~]$ ssh -A ec2-user@10.20.1.10
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Tue Oct 27 13:03:21 2026 from 10.10.2.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.137 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.197 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.177 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.176 ms

--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3145ms
rtt min/avg/max/mdev = 0.137/0.171/0.197/0.021 ms
```


rtt min/avg/max/mdev = 0.166/0.178/0.191/0.011 ms
[ec2-user@ip-10-10-2-10 ~]$ ssh -A ec2-user@10.20.1.10
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Tue Oct 27 13:03:21 2026 from 10.10.2.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10